Persona-Based Multi-Agent LLM Debate as an Automated Playtester: Validating Game Difficulty Data Against Human Performance Baselines

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CONTENTS

1 Introduction **1**

1.1 Background 1

1.2 Problem Statement 2

1.3 Research Questions 2

1.4 Research Objectives 3

1.5 Scope and Limitations 3

1.6 Expected Contributions 4

2 Literature Review **5**

2.1 Automated Game Playtesting 5

2.2 Game Difficulty and Behavioral Performance Measures 6

2.3 Related Work 7

2.4 Research Gap and Positioning 8

References **9**

CHAPTER 1

INTRODUCTION

1.1 Background

"Is the game hard enough for the intended player?" is one of the key questions that game designers keep asking during development [1]. A game that is too hard drives players away while one that is too easy soon becomes stale, both of which risk losing players and receiving a negative reception [2]. That is why difficulty can be an indicator of player engagement and experience [3, 4]. Designers measure this difficulty through playtesting. However, manual playtesting is resource-intensive since it needs many testers over many sessions and has to be repeated after every significant design change [1, 5, 6]. For indie game developers with limited budgets, this cost is often prohibitive. Thus, they need a cheaper way to obtain difficulty data that still reflects human players.

Automatic playtesting has gained popularity among industry and academic researchers alike. The methods explored so far fall into two groups. Hand-crafted methods, such as programmed test agents [5] and mathematical difficulty models [7], require heavy engineering effort for each game or genre. Learning-based methods such as deep reinforcement learning need large training resources while also tending to play in ways that humans would not [8]. Large Language Models (LLMs) provide an alternative, as LLM agents can play games ranging from chess [9] to Pokémon battles [10] and Minecraft [11] using only a text description of the rules. Xiao and Yang [12] show that the difficulty LLM agents experience in Wordle and Slay the Spire follows that of human players, even though the agents play worse. However, these LLM playtesters rely on a single agent that plays with only one playstyle. Human players are more diverse, as some take risks while others play safely. The difficulty that matters to a designer is the one experienced by this diverse population. Moreover, single LLMs replicate human strategic behavior less closely than multi-agent systems do [13]. Thus, it remains unclear whether an LLM playtester can produce difficulty data that is both consistent and aligned with human players once multiple playstyles are considered.

To accommodate this gap, we establish a persona-based multi-agent LLM debate framework that acts as an automated playtester. Such a playtester plays a game through the same actions available to a human player while recording the outcome of each playthrough. Game difficulty here refers to objective difficulty, which is the part of difficulty measured from play outcomes rather than the difficulty perceived by individual players [14]. In this framework, several LLM agents with different playstyle personas debate each in-game decision until they reach consensus on the action to play. In other words, the agents reach consensus on what to play rather than on how hard the game is. The difficulty data comes from the outcome of each playthrough, such

as the number of guesses used. These outcomes are then compared with human performance baselines, which are the average results of real players on the same puzzles. The exact agent roles and debate protocol are defined later in the research method.

We employ Wordle as the case study for three reasons. First, its games are short with deterministic feedback. Second, it was studied by Xiao and Yang [12], so our results can be related to prior single-agent work. Third, Wordle has public per-puzzle human results[1] that express the difficulty of each puzzle as the average number of guesses human players needed. The playtester plays a sample of historical puzzles several times each, after which we measure how much its outcomes vary across repeated runs of the same puzzle. We also measure how well its average guess counts agree with the human averages, both as continuous scores and as ordered difficulty categories. In addition, we describe how the playstyle personas behave during the debate, since a persona-based playtester is only meaningful when its personas actually shape the guesses.

1.2 Problem Statement

The background shows that game designers still lack an automated playtester whose outcomes can serve as evidence of game difficulty for human players. Existing LLM playtesters use a single agent with one playstyle [12], while the difficulty that matters to a designer is the one experienced by a diverse player population. For a persona-based playtester to fill this gap, two conditions must hold before its outcomes can be used as difficulty data. First, the outcomes must be consistent across runs. Since LLM outputs vary between calls, the same puzzle can end differently on different runs, which leaves a designer with little to rely on. Xiao and Yang [12] play each Wordle puzzle 20 times yet report only the average outcome, so this consistency is still unknown. Second, the outcomes must agree with human difficulty [12], because a consistent playtester that finds every puzzle equally easy is still of no use. Moreover, the playtester is only persona-based in practice if its personas keep their playstyles during the debate. Debating LLM agents may conform to their peers instead of keeping their assigned personas [15], which would leave a single playstyle behind the outcomes. This research therefore examines whether the proposed playtester meets both conditions in Wordle while describing how its personas behave.

1.3 Research Questions

Based on the problem statement, this research addresses the following research questions:

- **RQ1.** How consistent are the Wordle playtest outcomes produced by the persona-based multi-agent debate playtester across repeated runs on the same puzzles?

[1] Engaging Data, "Wordle Guess Distribution," `https://engaging-data.com/wordle-guess-distribution/`.

- **RQ2.** To what extent do the Wordle difficulty estimates derived from the playtester's outcomes agree with human difficulty, measured by average guess scores and difficulty categories?
- **RQ3.** How do the playstyle personas behave during the debate, and how strongly do their suggestions shape the final guesses?

Answering RQ1 and RQ2 shows whether the playtester can be trusted as a source of difficulty data for Wordle, which is the first step toward solving the general problem. RQ3 shows whether the personas actually take part in producing that data.

1.4 Research Objectives

The objectives of this research are fourfold:

1. To implement a persona-based multi-agent LLM debate playtester for Wordle, in which agents with different playstyle personas reach consensus on each guess while every playthrough is logged as difficulty data.
2. To measure the consistency of the playtester's outcomes across repeated runs of the same puzzles, which answers RQ1.
3. To measure the agreement between the playtester's difficulty estimates and human performance baselines using both average guess scores and difficulty categories, which answers RQ2.
4. To describe how each playstyle persona behaves during the debate and how often its suggestions are adopted in the final guesses, which answers RQ3.

1.5 Scope and Limitations

This research focuses on Wordle as a single case study. The playtester plays past Wordle puzzles for which public human results are available, so that every outcome has a human counterpart to compare with. Game difficulty is limited to objective difficulty, which is measured from how many guesses the playtester needs to solve each puzzle [14]. The playtester is evaluated as a whole system, judged by how consistent its outcomes are and how closely they agree with human players. The behavior of the personas is described from the debate records rather than tested by removing or swapping personas.

This research also has several limitations. It does not compare the debate playtester with a single-agent playtester [13]. Its persona analysis is also descriptive, so it shows how the personas take part in the debate without proving that any persona causes the outcomes. Moreover, the human baselines describe players as a group rather than as individuals, so the comparison can only be made for each puzzle as a whole. The difficulty perceived by players is not studied either

[14], since no human participants are involved. Finally, the findings apply to Wordle and to the LLM used in this research, while other games such as Slay the Spire [12] may behave differently.

1.6 Expected Contributions

This research is expected to contribute in four ways. First, it offers an automated playtester in which LLM agents with different playstyles decide together how to play a game. Second, it shows whether such a playtester gives similar results when it plays the same game again. Third, it shows how closely the difficulty found by the playtester follows the difficulty experienced by human players. Fourth, it shows how the playstyle personas take part in the debate, which reveals whether they actually shape the play.

For indie game developers, these results indicate whether an LLM playtester can provide difficulty data that reflects human players, which would reduce their reliance on manual playtesting [5, 6]. For researchers, this research extends earlier work on LLM playtesters [12] from a single agent to several agents with different playstyles [13].

CHAPTER 2

LITERATURE REVIEW

2.1 Automated Game Playtesting

Playtesting is conducted to observe how a game under development is experienced by its players. Through playtesting, designers can find faults in the game as well as design problems, such as levels that are harder than intended [1]. However, manual playtesting is a resource-intensive and rigorous process, since it needs many human testers over many sessions after every design change [5, 6]. Automatic playtesting addresses this cost by letting software agents play the game while recording each playthrough.

Various methods have been explored for automatic playtesting, ranging from hand-crafted agents to deep learning. Prasetya et al. [5] demonstrate a more "classical" yet highly powerful approach with the APLib framework. Through goal structures and domain-specific languages, they can create highly controlled test agents, although this requires significant programming effort with a steep learning curve. The mathematical model of Francillette et al. [7] for platformer games is also noteworthy. This model is highly deterministic and requires no sophisticated AI, yet it is limited to the platformer genre while demanding a complex setup.

Learning-based approaches reduce part of this manual work. Gudmundsson et al. [16] train a deep neural network on human moves in a puzzle game, after which the network predicts how hard new levels are for players. However, such an agent can only be used once a large set of human play data is available. Deep reinforcement learning agents learn from their own play instead, yet they demand substantial training resources. While these agents may outperform human players, they lack the natural behavioral patterns of actual human gameplay [8]. Holmgård et al. [17] narrow this gap with procedural personas, which are agents tuned to represent different playstyles. Their personas show that one playtester can cover several ways of playing, although each persona still relies on heuristics evolved for the specific game.

The emergence of LLMs provides an alternative approach for playtesting. LLM agents receive the game rules and the current game state as text, so they can play a new game without game-specific training [12]. Zhao and Tang [6] adopt this approach for match-3 games. They successfully address the inability of an LLM to "see" the game by converting the game board into numerical representations. As a result, their playtester achieves strong code coverage with effective bug detection. Bateni and Whitehead [18] show that LLM agents in Slay the Spire exhibit long-term planning without specialized training, which lets them adapt to changes in the game rules. Within difficulty measurement, Xiao and Yang [12] demonstrate that LLMs can indeed be employed to measure the difficulty of Wordle and Slay the Spire. Although their gameplay is inferior to that of human players, the difficulty patterns experienced by the

LLMs resemble those perceived by humans. Nevertheless, a single LLM loses too often in most games, which forces the authors to add a handicap system. Thus, LLMs offer a promising option for indie game developers, who rarely have the data or budget that learning-based approaches require.

2.2 Game Difficulty and Behavioral Performance Measures

A "fun" game can be linked to the design of its difficulty. Players may feel a sense of accomplishment when overcoming challenges, particularly when those challenges are significant [2]. For example, Dark Souls was dubbed the hardest game of its time, yet the higher the risk the player must take, the greater the reward [19]. The game garnered significant player engagement, which later gave rise to a new sub-genre known as "souls-like" [20]. Given the importance of difficulty, measuring it for the intended player is a key part of game design.

According to Constant et al. [14], there are two types of difficulty. First, subjective difficulty is how challenging players perceive the game to be, which cannot be measured explicitly because it is tied to personal skill, stress and emotion. Second, objective difficulty is the quantifiable aspect of a game that determines how challenging it is, independent of player perception. This research focuses on objective difficulty only, since an automated playtester produces play outcomes rather than perceptions of the game.

Objective difficulty is measured through behavioral performance measures, which record what players do and achieve while playing. Such measures include whether a challenge is completed, how many attempts it takes and how long it takes. Dynamic difficulty adjustment systems rely on these measures to estimate the skill of a player during play [1, 3]. Playtesting studies use them to compare agents with humans, for instance the level success rate of an agent against the pass rate of human players [16].

In Wordle, the difficulty is operationalized through the guess count, which is how many attempts a player requires to identify the target word. Lower guess counts indicate easier words, while higher counts or a lost game indicate harder ones. Engaging Data[1] provides the guess distribution of real players for each past puzzle, from which the average number of guesses per puzzle can be computed. For example, a word with an average of 3.2 guesses is objectively easier than one with 5.8 guesses. However, the dataset does not provide additional details such as demographic information about the players. This average serves as the human performance baseline in this research, as it does in the Wordle comparison of Xiao and Yang [12].

Comparing a playtester with human players raises two separate questions. The first is whether the playtester gives similar outcomes on the same puzzle again, since a single run of an LLM agent may not represent its typical play. This reliability can be assessed through outcome variance, such as the standard deviation of repeated playthroughs. The second question is whether

[1] Engaging Data, "Wordle Guess Distribution," `https://engaging-data.com/wordle-guess-distribution/`.

the playtester ranks and groups puzzles in the same way as human results do. Agreement can be checked on continuous scores, such as the correlation between average guess counts, or on ordered categories from easy to hard. Since the human baselines describe the player population rather than individual players, both comparisons are made per puzzle over a random sample of puzzles.

2.3 Related Work

Recent studies demonstrate that LLM agents can play diverse games. ChessGPT bridges policy learning and language modeling for chess [9], while PokeLLMon reaches human-parity performance in Pokémon battles [10]. Voyager acts as an embodied agent in Minecraft that keeps learning new skills as it explores [11]. However, these agents aim to play well rather than to play like humans, which differs from the goal of playtesting.

For playtesting, the main concern is whether the results of an agent reflect human players. The LLM playtesters reviewed above play alone with a single playstyle [6, 12], so they do not account for the diversity of player playstyles. Sreedhar and Chilton [13] show in a strategic game that multi-agent systems replicate human behavior more closely than single LLMs do. Before LLMs, Holmgård et al. [17] already represented different players through procedural personas. More recently, MIMIC [21] equips an LLM game tester with different personality traits, which leads to more diverse solutions with higher test coverage. Thus, capturing the diversity of players requires moving beyond a single LLM agent.

Multi-agent debate is one way to combine several LLM agents into a single decision. In a debate, agents propose their own answers before reading the reasoning of the others, after which they revise their answers over several rounds until they agree [22]. Du et al. [22] show that this process improves factual accuracy and reasoning compared with a single agent. Liang et al. [23] let agents express arguments in a state of "tit for tat", where each agent directly responds to the position of the previous agent. Meanwhile, a managing agent oversees the exchange to obtain the final decision. ChatEval [24] assigns different roles to agents in a debate for evaluating text, while finding that the diversity of these roles is important for its results. However, Smit et al. [25] find that debate does not reliably outperform other prompting strategies, as its results depend strongly on how the debate is set up. Personas also do not always hold in a debate, since agents may conform to their peers or drift away from their assigned views [15].

We adopt the debate structure of Liang et al. [23] for two reasons. First, its managing agent provides a single point where the arguments end in one action, which a playtester needs because every turn must produce a playable move. Second, agents that argue from different positions keep the debate from reaching agreement immediately, which matches the role of playstyle personas [24]. Since the benefit of debate depends on its setup [25], we evaluate the resulting playtester directly rather than assuming that debate improves it.

2.4 Research Gap and Positioning

The reviewed studies reveal a trade-off between generality and human likeness. Hand-crafted playtesters are highly controlled but must be rebuilt for each game [5, 7]. Learning-based playtesters require either large sets of human play data or substantial training resources [8, 16]. LLM playtesters can play a new game from its rules alone, while the difficulty they experience can follow that of human players [12]. However, existing LLM playtesters rely on a single agent with one playstyle, even though personas have long been used to represent different players [17].

While multi-agent debate could reconcile these heterogeneous playstyles into one consensus action, automated consensus frameworks remain absent from current difficulty measurement methods. Thus, three questions remain open. First, LLM outputs vary between calls, so it is unclear whether a playtester built from several debating agents produces consistent outcomes across repeated runs. Even Xiao and Yang [12] play each Wordle puzzle 20 times yet report only the average outcome. Second, it is unclear whether the difficulty found by such a playtester still agrees with human performance baselines. Third, it is unclear whether the personas keep their playstyles during the debate, since debating agents may conform to their peers [15]. To the best of our knowledge, no prior study has applied persona-based multi-agent debate to measure game difficulty.

To accommodate this gap, our research is positioned where these lines of work meet. We take playstyle personas from persona-based playtesting [17] while taking the consensus process from multi-agent debate [23]. Both are then applied to the LLM difficulty measurement of Xiao and Yang [12], using Wordle and its public human results. Unlike studies that compare multi-agent systems with a single agent [13, 25], this research validates the playtester as a system. It does so through the consistency of its outcomes (RQ1), their agreement with human difficulty (RQ2) and the behavior of its personas during the debate (RQ3).

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